DATA SCIENCE CASE NOTES IBM/CognitiveClass

Data Science 101 *provides a foundational overview of the field, defining it as the disciplined process of extracting actionable insights from data to satisfy human curiosity and address critical business imperatives. It establishes how leveraging data science enables organizations to make evidence-based decisions, thereby enhancing performance and competitive advantage. The module introduces the fundamental dichotomy of data types—structured versus unstructured—and emphasizes that systematic data collection is a prerequisite for any meaningful analysis. Finally, it outlines the transformative role of cloud computing in modern data science, explaining how the cloud serves as a scalable platform that overcomes traditional limitations of physical infrastructure, augments storage and compute capacity, and streamlines collaborative workflows for data teams.*

For example, a retail chain uses data science to analyze structured sales data (transaction records) and unstructured social media sentiment (customer reviews) to understand why a product is underperforming. By collecting and analyzing this data in a cloud environment like AWS, their data science team can rapidly process terabytes of information without investing in physical servers, ultimately generating insights that lead to a targeted marketing campaign, boosting sales by 15%.

The Data Science Methodology *provides a structured, iterative framework for solving data-driven problems, emphasizing that a clear business understanding must precede any technical solution. This initial phase defines the problem and determines the relevant data needed. It then guides practitioners through a logical sequence: first, selecting an appropriate analytic approach (e.g., descriptive, predictive); second, specifying detailed data requirements and sourcing; and third, entering a cycle of Data Understanding and Preparation. Data Understanding involves exploring data via descriptive statistics and*

assessing quality, while Data Preparation—often the most time-intensive phase—cleans, transforms, and formats data for modeling. The Modeling phase then selects and applies suitable algorithms to build the solution. Finally, the Evaluation phase rigorously tests model performance, leading to Deployment for real-world use and impact.

A clear application of the methodology is the healthcare case study predicting whether a heart failure patient will be readmitted within 30 days. It begins with the business objective of reducing costs and improving care. The team frames it as a predictive classification problem, sourcing clinical data (e.g., ejection fraction, creatinine). After exploring and cleaning this data, they train classification models like Random Forest. Models are evaluated prioritizing high recall to identify most at-risk patients, and the final model is deployed to flag high-risk cases in hospital discharge systems, enabling proactive intervention.

The Data Science Tools and Workflow modules establish the essential, layered toolkit required to operationalize the data science methodology. This begins with the foundational languages: Python, as the dominant language for general-purpose analysis and machine learning; R, specialized for statistical computing; and SQL, the universal language for data extraction and manipulation. The power of these languages is unlocked through their ecosystems of libraries and frameworks—such as Pandas for data wrangling, NumPy for numerical computation, and Scikit-learn for modeling—which provide standardized, efficient implementations of complex algorithms. Critically, the work is conducted within integrated development environments like Jupyter Notebooks and RStudio, which combine code execution, visualization, and documentation in interactive, reproducible formats. The curriculum further contrasts open-source tools with commercial and cloud-based platforms (e.g., IBM Watson Studio), which provide managed, scalable infrastructure for collaborative projects. Finally, the toolchain is completed by version control systems, primarily Git and GitHub, which are fundamental for managing code versions, enabling team

collaboration, and ensuring project reproducibility from exploration to deployment.

For example, a financial analyst forecasting quarterly revenue would first use SQL to query the company's data warehouse for historical sales and marketing data. They would then open a Jupyter Notebook in their company's Watson Studio cloud environment, using Python with Pandas to clean the data and create features like 'quarter-over-quarter growth.' They might use Scikit-learn to build a time series model, sharing the notebook via GitHub for their manager's review. The final model could be deployed as a REST API from the same platform, updating forecasts automatically each week. This workflow leverages the full stack: languages, libraries, a collaborative environment, and a cloud platform for deployment

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